SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design **Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:50
Duration : 1 Hour

Scenario Based

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. A company is developing a smart security system for a research laboratory. Every employee entering the lab uses a digital access card that generates a sequence of symbols:

- a → Valid ID scan
- b → Biometric verification
- c → Door access confirmation

The security rule states that:

1. Every valid access sequence must start with a.
2. It may contain any number of biometric checks (b).
3. The sequence must end with c.

Examples of valid sequences: ac, abc, abbc, abbbc

